

SCI225 Week 3 — Complete Study Guide

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Biology of Cancer · Cancer Epidemiology · Pediatric & Adolescent Malignancies

Study Spine — for every cancer ask:

1. Which gene category drives it (oncogene, tumor suppressor, DNA repair)? 2. Tissue of origin? 3. Stage vs grade? 4. Modifiable risk factor or hereditary?

MECHANISM

WHY

COMPENSATION

PATHOLOGY

MISSTEP

ACRONYM

EXAM TIP

Course alignment

- **LO1** — Recall the Biology of Cancer and specifically how it occurs (Units 1–3)
- **LO2** — Define and discuss Cancer Epidemiology and the specific consequences that it entails (Unit 4)
- **LO3** — Identify different types of cancer in children and adolescents (Units 5, 6)

Every practice question here was written to teach you to apply the concept. None reproduces an exam item.

Unit 1 • Biology of Cancer LO1

MECHANISM Cancer is the outcome of **accumulated mutations** in a cell that disable the normal controls on growth, death, and tissue boundaries. The result: a clone of cells that proliferates, refuses to die, evades the immune system, builds its own blood supply, invades adjacent tissue, and ultimately spreads.

1.1 Three Gene Categories That Matter in Cancer

Gene type	Normal job	Cancer change	How many "hits" needed?	Examples
Proto-oncogene → oncogene	Promote cell growth (gas pedal)	Activated / overexpressed → uncontrolled growth	Just one allele needs to be activated (gain of function — dominant)	HER2 (breast), RAS (many), MYC (lymphoma), BCR-ABL (CML)
Tumor suppressor gene	Brake on cell growth (brake pedal)	Inactivated / lost → no brakes	Both alleles must be lost (Knudson's two-hit hypothesis — recessive at the cell level)	TP53 (~half of all cancers), RB1 (retinoblastoma), BRCA1/2, APC (colon)
DNA repair gene	Fix mutations	Loss of repair → mutations	Both alleles (similar logic to suppressors)	BRCA1/2, MLH1/MSH2 (Lynch syndrome)

	as they happen	accumulate elsewhere		
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WHY Why oncogenes need only one hit but tumor suppressors need two.

A proto-oncogene normally has carefully regulated activity. Activate *one* copy abnormally and it overproduces growth signals — the other normal copy can't shut it off. Gain of function = dominant.

A tumor suppressor still works if *one* copy is intact (the other does the job). You need to lose **both** copies before the brake is gone — Knudson's two-hit. Loss of function = recessive at the cellular level.

How oncogenes get activated

- **Point mutation** — single nucleotide change makes the gene constitutively active (RAS)
- **Gene amplification** — extra copies of the gene (HER2 amplified in some breast cancers)
- **Chromosomal translocation** — gene is moved next to an active promoter (BCR-ABL Philadelphia chromosome in CML)
- **Insertion / viral integration** — virus inserts near the gene (HTLV-1, HPV)

★ **EXAM TIP HER2 overexpression in breast cancer** is a textbook oncogene-activation example. The HER2 gene is amplified, the receptor is overexpressed on the cell surface, and the tumor responds to anti-HER2 therapy (trastuzumab). When you see "HER2 amplified" or "HER2 positive," think oncogene activation by amplification.

1.2 Stages of Cancer Development

Normal → hyperplasia → dysplasia → carcinoma in situ → invasive carcinoma → metastatic carcinoma

Stage	What's happening
Hyperplasia	Increased number of normal-looking cells in response to a stimulus. Reversible if stimulus removed.
Dysplasia	Disorganized, abnormal-looking cells; loss of normal architecture; nuclear pleomorphism (size/shape variation), increased nuclear-to-cytoplasm ratio. Still confined to the tissue. Often reversible if cause removed.
Carcinoma in situ	Severely dysplastic cells fill the full thickness of the epithelium but have NOT yet broken through the basement membrane. "Pre-invasive cancer."
Invasive carcinoma	Tumor cells breach the basement membrane and invade surrounding stroma. Capable of metastasis from this point.
Metastatic disease	Tumor cells reach distant sites via lymphatic or hematogenous spread.

MECHANISM What the basement membrane crossing means. Once tumor cells break through the basement membrane, they have access to lymphatics and blood vessels. **Crossing the basement membrane is the defining event that converts pre-invasive dysplasia/CIS into "real" cancer** — and it requires acquiring the ability to make matrix-degrading enzymes (matrix metalloproteinases).

⚠ **COMMON MISSTEP** "Dysplasia is cancer." Not yet — dysplasia is pre-cancerous, often reversible. **Carcinoma in situ** is the immediate precursor. **Invasive carcinoma** is the first stage that meets the definition of cancer (basement membrane crossed).

Practice Questions — Biology of Cancer

Q1. Select all that apply. Which findings are consistent with oncogene activation?

- A. HER2 receptor overexpression in breast cancer
- B. RAS gene point mutation producing constitutive growth signaling
- C. Both alleles of TP53 inactivated
- D. BCR-ABL translocation in chronic myeloid leukemia
- E. Loss of function of BRCA1

[Show answer](#)

A, B, D. All gain-of-function changes in proto-oncogenes. C (TP53 loss) and E (BRCA1 loss) are tumor-suppressor inactivations — the opposite mechanism, requiring both alleles to be lost.

Q2. Matching. Match each gene category to its function.

Gene type	Function
Proto-oncogene	_____
Tumor suppressor	_____
DNA repair gene	_____

Choices: Repairs DNA damage; loss → mutation accumulation · Promotes cell proliferation; gain of function → uncontrolled growth · Inhibits cell proliferation; loss of function → no brakes

[Show answer](#)

Proto-oncogene → promotes proliferation, gain of function = uncontrolled growth · Tumor suppressor → inhibits proliferation, loss of function = no brakes · DNA repair gene → repairs damage, loss = mutation accumulation.

Q3. Order. Place the stages of cancer development in correct order:

- Carcinoma in situ
- Invasive carcinoma
- Hyperplasia
- Metastatic carcinoma
- Dysplasia

[Show answer](#)

Hyperplasia → dysplasia → carcinoma in situ → invasive carcinoma → metastatic.
Crossing the basement membrane separates CIS from invasive disease.

Q4. A pancreatic biopsy shows disorganized cells with marked variation in nuclear size and shape. The tumor has not breached the basement membrane. Which best describes this finding?

- A. Hyperplasia
- B. Dysplasia / carcinoma in situ
- C. Invasive adenocarcinoma
- D. Metastatic disease

[Show answer](#)

B. Disorganized cells with nuclear pleomorphism = dysplasia. If those cells span the full epithelium without basement-membrane breach, that's CIS. Basement-membrane crossing is the gateway to "invasive."

Q5. Select all that apply. Which mechanisms can activate a proto-oncogene?

- A. Point mutation producing a constitutively active protein
- B. Gene amplification (extra copies)
- C. Chromosomal translocation that places the gene under an active promoter
- D. Loss of both alleles
- E. Inability to repair DNA damage

[Show answer](#)

A, B, C. D (loss of both alleles) is the mechanism for tumor suppressor inactivation, not oncogene activation. E describes a DNA repair gene defect.

Unit 2 • Tumor Behavior — Spread, Angiogenesis, Cachexia LO1

2.1 Invasion and Metastasis

Tumor at primary site → degrades basement membrane (matrix metalloproteinases) → invades local stroma → enters lymphatic or blood vessel → survives in circulation → arrests in distant capillary bed → extravasates into new tissue → grows new colony at distant site

MECHANISM Metastasis = the ability to seed new tumors in distant tissues. This is the property that makes cancer lethal. Without it, even large primary tumors can often be removed surgically. Cells must acquire all of the following: ability to detach from neighbors, degrade matrix, intravasate into vessels, survive shear forces and immune attack, and proliferate in foreign tissue.

Route	Typical for	Common destinations
Lymphatic	Most carcinomas	Regional lymph nodes first, then central nodes
Hematogenous (blood)	Sarcomas; many late-stage carcinomas	Liver, lung, bone, brain — the "big four" metastatic sites
Direct extension	Most tumors	Adjacent organs and structures
Seeding	Tumors of body cavities (ovarian, gastric)	Peritoneum, pleura

★ **EXAM TIP** "Which characteristic of tumor cells directly allows spread to distant organs?" The answer is the ability to **invade through the basement membrane and enter blood or lymphatic vessels**. Without invasion, there is no metastasis.

2.2 Angiogenesis

MECHANISM A tumor larger than ~1–2 mm cannot survive on diffusion alone — it must build new blood vessels (angiogenesis) to deliver oxygen and nutrients and remove waste. Tumors secrete **vascular endothelial growth factor (VEGF)** and other angiogenic factors that recruit endothelial cells from existing capillaries to form new vessels.

WHY Why anti-angiogenic drugs work in cancer. Drugs like bevacizumab (anti-VEGF) starve tumors by blocking the new-vessel signal. Tumors cannot grow past the diffusion limit without their own vasculature, and without growth, metastasis is also limited.

2.3 Cancer Cachexia and Paraneoplastic Syndromes

PATHOLOGY Cancer cachexia = the wasting syndrome of advanced cancer. Severe weight loss, muscle loss, fatigue, and weakness *despite adequate caloric intake*. Driven by tumor- and host-derived **inflammatory cytokines** — TNF- α , IL-1, IL-6 — which:

- Suppress appetite (anorexia)
- Increase resting metabolic rate
- Promote skeletal muscle protein breakdown (proteolysis)
- Promote lipid breakdown (lipolysis)
- Cause insulin resistance

The tumor itself also competes for nutrients, but the cytokine-driven hypermetabolism is the key reason eating more does not fix the wasting.

WHY Why patients lose weight despite eating. The body is in a *catabolic* state — breaking down muscle and fat faster than food can replace. Caloric supplementation alone cannot overcome the inflammatory drive. Best management requires treating the underlying cancer plus supportive nutrition.

Paraneoplastic syndromes — tumor effects unrelated to direct invasion

Syndrome	Cancer	Mechanism
SIADH (low Na, concentrated urine)	Small-cell lung cancer	Ectopic ADH production
Cushing syndrome	Small-cell lung cancer	Ectopic ACTH production
Hypercalcemia	Squamous cell lung cancer, multiple myeloma, breast cancer with bone mets	PTH-related peptide; bone destruction
Lambert-Eaton syndrome	Small-cell lung cancer	Antibodies vs presynaptic Ca ²⁺ channel at NMJ
Polycythemia	Renal cell carcinoma	Ectopic erythropoietin

★ **EXAM TIP "Bystander mutations"** — a melanoma sample with many mutations that don't change tumor behavior is showing **passenger mutations**. Cancers accumulate

hundreds of incidental mutations alongside the few "driver" mutations that actually fuel growth. Recognizing this distinction matters for understanding tumor genomics.

Practice Questions — Tumor Behavior

Q6. Which characteristic of tumor cells most directly allows spread to distant organs?

- A. Increased size of tumor cells
- B. Ability to invade through the basement membrane and enter blood or lymphatic vessels
- C. Slower-than-normal cell division
- D. Increased oxygen consumption

[Show answer](#)

B. Crossing the basement membrane and entering vessels is the gateway to metastasis. Without invasion, tumors stay local.

Q7. A lung tumor shows extensive new blood vessel formation on biopsy. Which process most directly accounts for this?

- A. Apoptosis
- B. Angiogenesis driven by tumor-released VEGF
- C. Necrosis of stromal cells
- D. Inflammatory edema

[Show answer](#)

B. VEGF and other tumor-secreted angiogenic factors recruit endothelial cells to build new vessels — without them, tumors cannot grow past 1–2 mm.

Q8. Select all that apply. A patient with metastatic breast cancer has weight loss, muscle loss, and high inflammatory markers. Which mechanisms explain cachexia in this patient?

- A. TNF- α , IL-1, and IL-6 drive proteolysis and lipolysis
- B. Hypermetabolism increases caloric expenditure
- C. Cytokines suppress appetite
- D. The tumor itself uses energy and nutrients
- E. The patient is simply not eating enough food

[Show answer](#)

A, B, C, D. E is incorrect — patients with cachexia often eat normally; cytokine-driven catabolism is the dominant mechanism. Just adding calories is not sufficient.

Q9. A breast biopsy shows marked HER2 overexpression. Which gene mechanism is responsible?

- A. Loss of both copies of a tumor suppressor
- B. Gain of function in a proto-oncogene through gene amplification
- C. Mutation of a DNA repair gene
- D. Translocation of HER2 onto chromosome 22

[Show answer](#)

B. HER2 amplification produces extra copies of the gene → overexpression of the receptor → unchecked growth signaling. Targeted therapy (trastuzumab) blocks this receptor.

Unit 3 • Naming, Grading & Staging LO1

3.1 Naming Conventions

Tissue of origin	Benign	Malignant
Epithelium	-oma (e.g., adenoma, papilloma)	-carcinoma (adenocarcinoma, squamous cell carcinoma)
Connective tissue (bone, muscle, fat, cartilage)	-oma (lipoma, chondroma, leiomyoma)	-sarcoma (osteosarcoma, liposarcoma, rhabdomyosarcoma)
Hematopoietic (blood-forming)	—	Leukemia (in blood/marrow); lymphoma (in lymph nodes/tissue)
Embryonic / "blast" cells	—	-blastoma (neuroblastoma, retinoblastoma, hepatoblastoma)

⚠ **COMMON MISSTEP** "All -omas are benign." Watch out for these malignancies that end in -oma despite being malignant: **lymphoma, melanoma, hepatoma (hepatocellular carcinoma), seminoma**. Memorize this short list.

3.2 Grading vs Staging — Don't Confuse Them

	Grading	Staging
What it measures	How <i>differentiated</i> (how much the cells resemble normal tissue)	How far the cancer has <i>spread</i>
Source	Microscopic appearance (pathologist looking at biopsy)	Clinical exam, imaging, surgical findings
Scale	Grade I (well-differentiated, looks like normal) → Grade IV (poorly differentiated, anaplastic)	TNM: T umor size · N odes · M etastasis. Combined into stages 0–IV.
What it predicts	How aggressive the tumor cells are intrinsically	Prognosis and treatment plan (usually predicts outcome better than grade)

ACRONYM / MNEMONIC "Stage = Spread, Grade = Grossness."

Stage answers "where is it?" Grade answers "how ugly is it under the microscope?"

★ **EXAM TIP** For most cancers, **stage predicts prognosis better than grade**. A small, low-stage but high-grade tumor often has a better outcome than a large, widely metastatic but low-grade tumor.

Practice Questions — Naming, Grading, Staging

Q10. Matching. Match each tumor type to its description.

Tumor name	Description
Adenocarcinoma	—

Osteosarcoma	_____
Leukemia	_____
Lipoma	_____

Choices: Benign tumor of fat tissue · Malignant blood-forming cell cancer in marrow · Malignant epithelial gland tumor · Malignant bone tumor

[Show answer](#)

Adenocarcinoma → malignant epithelial gland tumor · Osteosarcoma → malignant bone tumor · Leukemia → malignant blood-forming cell cancer in marrow · Lipoma → benign fat tumor.

Q11. A patient has a 5-cm pulmonary mass with regional lymph nodes involved but no distant metastasis. Which staging element captures the lymph involvement?

- A. T (tumor size)
- B. N (regional nodes)
- C. M (distant metastasis)
- D. Grade

[Show answer](#)

B. TNM: T = primary tumor size and extent, N = regional lymph node involvement, M = distant metastasis. Grade is a separate measure (differentiation), not part of TNM.

Unit 4 • Cancer Epidemiology — Risk Factors & Prevention LO2

4.1 Modifiable Risk Factors

Exposure	Cancers most associated	Mechanism
Tobacco / smoking	Lung, larynx, oral, esophagus, bladder, kidney, pancreas, cervix	Carcinogens in smoke (nitrosamines, polycyclic aromatic hydrocarbons) cause DNA damage; chronic inflammation; metaplasia → dysplasia → cancer
Chronic alcohol use	Oral cavity, pharynx, larynx, esophagus, liver, breast, colorectal	Alcohol metabolism produces acetaldehyde — a highly toxic, carcinogenic metabolite that damages DNA
Obesity / metabolic syndrome	Endometrial, postmenopausal breast, colorectal, kidney, esophageal adenocarcinoma, pancreatic, liver	Chronic inflammation, insulin resistance, hyperinsulinemia, ↑ estrogen from adipose tissue
UV radiation (sun, tanning beds)	Basal cell, squamous cell, melanoma	UV-induced pyrimidine dimers in skin DNA → mutations in TP53 and other genes; cumulative damage with age
Ionizing radiation	Leukemia, thyroid, breast	Direct DNA strand breaks

HPV infection	Cervical, anal, oropharyngeal	Viral E6 inactivates p53; E7 inactivates Rb (tumor suppressors)
Hepatitis B/C	Hepatocellular carcinoma	Chronic inflammation → cirrhosis → cancer
H. pylori	Gastric, MALT lymphoma	Chronic gastritis → atrophy → metaplasia → dysplasia → cancer
Diethylstilbestrol (DES)	Clear cell adenocarcinoma of the vagina/cervix in daughters of mothers who took DES	Transplacental hormonal disruption of müllerian development

WHY Why smoking causes lung cancer. Each cigarette delivers thousands of carcinogens that damage bronchial DNA. The bronchial epithelium first undergoes metaplasia (ciliated columnar → squamous), then dysplasia, then carcinoma in situ, and finally invasive cancer. The progression takes years. Stopping smoking reduces risk over time but never returns it to the never-smoker baseline.

WHY Why obesity increases cancer risk. Adipose tissue is metabolically active — it produces inflammatory cytokines (IL-6, TNF-α), aromatizes androgens to estrogen (raising postmenopausal breast and endometrial cancer risk), and drives insulin resistance with hyperinsulinemia (a growth signal for tumor cells). Metabolic syndrome links these together: central obesity, hypertension, dyslipidemia, hyperglycemia, hyperinsulinemia.

PATHOLOGY Alcohol and acetaldehyde. Ethanol → alcohol dehydrogenase → **acetaldehyde** → aldehyde dehydrogenase → acetate. Acetaldehyde is the carcinogenic intermediate. Patients with reduced ALDH activity (common in some Asian populations) accumulate more acetaldehyde and have higher cancer risk per drink.

4.2 Hereditary Cancer Syndromes

Syndrome	Gene	Major cancers
Hereditary breast/ovarian	BRCA1, BRCA2	Breast, ovarian, prostate, pancreatic
Lynch syndrome (HNPCC)	MLH1, MSH2 (DNA mismatch repair)	Colorectal, endometrial, ovarian, gastric
Familial adenomatous polyposis (FAP)	APC	Colorectal (100s of polyps; nearly 100% cancer risk by age 40 if untreated)
Li-Fraumeni	TP53	Many — sarcomas, breast, brain, leukemia, adrenal
Retinoblastoma	RB1	Retinoblastoma, osteosarcoma
MEN syndromes	RET (MEN2), MEN1	Endocrine tumors

★ **EXAM TIP "Mother and sister with breast cancer"** stems should make you think **BRCA1/BRCA2**. These are autosomal dominant; first-degree relatives of carriers should be offered genetic counseling. Risk-reducing strategies include enhanced screening, prophylactic mastectomy, oophorectomy, and chemoprevention.

Practice Questions — Epidemiology

Q12. Select all that apply. Which exposures increase cancer risk?

- A. Long-term tobacco smoking
- B. Chronic alcohol use
- C. Daily moderate exercise
- D. Cumulative UV radiation exposure
- E. Chronic HPV infection
- F. Obesity with metabolic syndrome

[Show answer](#)

A, B, D, E, F. C is protective. The other five are major modifiable cancer risk factors.

Q13. Why does long-term sun exposure increase the risk of skin cancer?

- A. Sunlight increases vitamin D, which is carcinogenic
- B. UV radiation creates pyrimidine dimers in skin DNA, leading to mutations in TP53 and other genes
- C. Sunlight directly burns away tumor suppressor genes
- D. UV radiation increases skin pigmentation, which is itself cancer

[Show answer](#)

B. UVB radiation produces pyrimidine dimers in DNA, which if not repaired produce mutations including in TP53. Cumulative damage drives basal cell, squamous cell, and melanoma development.

Q14. A patient with long-term alcohol use develops liver disease and is told she is at higher cancer risk. Which highly toxic metabolite contributes most directly?

- A. Methanol
- B. Acetaldehyde
- C. Glucose
- D. Lactate

[Show answer](#)

B. Alcohol → acetaldehyde → acetate. Acetaldehyde is the carcinogenic intermediate that damages DNA in the liver, oral cavity, esophagus, and beyond.

Q15. A 24-year-old presents with abnormal vaginal bleeding. Her mother took DES during pregnancy. Which cancer is she at increased risk for?

- A. Squamous cell carcinoma of the cervix
- B. Clear cell adenocarcinoma of the vagina or cervix
- C. Endometrial adenocarcinoma
- D. Ovarian serous carcinoma

[Show answer](#)

B. Daughters of mothers who took DES (an estrogen used to prevent miscarriage from the 1940s–70s) have an elevated risk of clear cell adenocarcinoma of the vagina/cervix — a classic transplacental carcinogen example.

Q16. A patient with central obesity, elevated fasting glucose, elevated fasting insulin, and increased waist circumference is at increased risk for which cancers?

- A. Breast (premenopausal only)
- B. Endometrial, postmenopausal breast, colorectal, and kidney cancers
- C. Acute leukemia
- D. None — metabolic syndrome is purely cardiovascular

[Show answer](#)

B. Metabolic syndrome and obesity drive multiple cancers via inflammation, insulin/IGF signaling, and aromatase-mediated estrogen excess.

Q17. A patient has a mother and sister with breast cancer. The most appropriate genetic concern is:

- A. Trisomy 21
- B. BRCA1 or BRCA2 mutation
- C. Klinefelter syndrome
- D. Lynch syndrome

[Show answer](#)

B. Two first-degree relatives with breast cancer should prompt BRCA testing referral. BRCA1/2 mutations are autosomal dominant tumor-suppressor losses associated with breast and ovarian cancer (and others).

Unit 5 • Pediatric Cancers LO3

MECHANISM Childhood cancers are biologically different from adult cancers. They more often arise from **embryonal tissue** ("-blastomas"), are more often associated with germline genetic predisposition, and tend to involve the blood, brain, and abdominal organs more than the epithelial sites that drive adult cancers.

WHY **Why leukemia is the most common childhood cancer.** Children's bone marrow is highly proliferative — building red cells, white cells, and platelets at high rates throughout growth. Highly proliferating tissues are more vulnerable to mutations during DNA replication. Add limited exposure to environmental carcinogens that drive adult epithelial cancers, and the result is a disease pattern dominated by hematopoietic malignancies.

5.1 Acute Lymphoblastic Leukemia (ALL)

PATHOLOGY ALL is the most common childhood cancer. Peak age 2–5. Driven by accumulation of immature lymphoblasts in the marrow that crowd out normal hematopoiesis.

Classic presentation:

- **Anemia** → fatigue, pallor
- **Thrombocytopenia** → bruising, petechiae, mucosal bleeding
- **Neutropenia** → frequent or severe infections, fever
- **Lymphadenopathy and hepatosplenomegaly**
- Bone pain (especially limp or refusal to walk)
- Lab: high WBC with many **blasts**, low Hgb, low platelets

★ **EXAM TIP** "4-year-old with fatigue, fever, frequent infections, pallor, and petechiae + anemia, thrombocytopenia, and increased immature WBCs" = ALL, classic stem.

5.2 CNS Tumors — Second Most Common Childhood Cancer

MECHANISM In contrast to adults (where most brain tumors are supratentorial), most childhood brain tumors arise **infratentorially** — in the cerebellum or brainstem. The most common pediatric CNS tumor is **medulloblastoma** of the cerebellum.

Classic presentation of pediatric posterior fossa tumor:

- Morning headaches, vomiting (especially on waking)
- **Cerebellar signs** — unsteady gait, ataxia, dysmetria, intention tremor
- Obstructive hydrocephalus from 4th ventricle compression — papilledema, increased head circumference in infants
- Cranial nerve palsies if brainstem involvement

5.3 Embryonal Tumors — The "-blastomas"

Tumor	Origin / location	Classic age	Hallmark presentation
Wilms tumor (nephroblastoma)	Kidney	2–5 years	Painless unilateral abdominal mass , sometimes with hematuria; the tumor stays confined to the kidney early
Neuroblastoma	Adrenal medulla / sympathetic chain	< 5 years (often < 2)	Abdominal mass with calcifications , irritability, hypertension, may cross the midline; high urinary catecholamine metabolites (HVA, VMA)
Retinoblastoma	Retina (RB1 gene)	< 5 years	Leukocoria ("white pupil reflex"), strabismus; can be hereditary (germline RB1 mutation, often bilateral) or sporadic
Hepatoblastoma	Liver	< 3 years	Abdominal mass, ↑ AFP

★ **EXAM TIP** The abdominal mass distinction:

Wilms = kidney, unilateral, does NOT cross midline.

Neuroblastoma = adrenal/sympathetic, often crosses midline, calcifications on imaging, urinary HVA/VMA elevated.

This pair shows up on every pediatric oncology test.

ACRONYM / MNEMONIC "**Leukocoria = retinoblastoma.**" White pupil reflex on a flash photograph in a young child is retinoblastoma until proven otherwise. Examine the other eye too — bilateral disease suggests a germline RB1 mutation and additional cancer risk later in life.

PATHOLOGY Common features of childhood cancer compared to adult cancer:

- More often arise from embryonal tissues
- More often associated with germline predisposition
- Generally **more chemo-sensitive** and curable
- Less linked to environmental carcinogen exposure

- Lymphoma, leukemia, and CNS tumors dominate the case list

Practice Questions — Pediatric Cancers

Q18. A 4-year-old has fatigue, fever, frequent infections, pallor, petechiae. CBC shows anemia, thrombocytopenia, and increased immature white blood cells. Most likely diagnosis?

- A. Wilms tumor
- B. Neuroblastoma
- C. Acute lymphoblastic leukemia (ALL)
- D. Medulloblastoma

[Show answer](#)

C. Marrow infiltration by lymphoblasts → pancytopenia + circulating blasts → anemia (fatigue, pallor), thrombocytopenia (petechiae), neutropenia (infection). Classic ALL.

Q19. A 3-year-old presents with a firm unilateral abdominal mass and hematuria. Imaging shows the tumor is confined to the kidney. Most likely diagnosis?

- A. Neuroblastoma
- B. Wilms tumor (nephroblastoma)
- C. Hepatoblastoma
- D. Lymphoma

[Show answer](#)

B. Painless unilateral abdominal mass + hematuria + tumor confined to kidney = Wilms tumor. Does not cross the midline.

Q20. A 2-year-old has abdominal swelling and irritability. Imaging shows an adrenal mass with calcifications. Best diagnosis?

- A. Wilms tumor
- B. Neuroblastoma
- C. Medulloblastoma
- D. Retinoblastoma

[Show answer](#)

B. Adrenal location + calcifications + young child = neuroblastoma. May cross midline; urinary catecholamine metabolites (HVA, VMA) elevated.

Q21. Dropdown. A child has a white pupil reflection (leukocoria). The tumor is [select: retinoblastoma / neuroblastoma / glioblastoma], which originates from [select: retinal cells / sympathetic ganglia / glial cells].

[Show answer](#)

Retinoblastoma · retinal cells. Leukocoria is the cardinal sign. The RB1 gene is the prototypical tumor suppressor (Knudson two-hit hypothesis).

Q22. A child has morning headaches, vomiting, unsteady gait, and poor coordination. Imaging shows a posterior fossa mass. Best diagnosis?

- A. Wilms tumor
- B. Cerebellar tumor (most commonly medulloblastoma)

- C. Neuroblastoma
- D. Acute leukemia

[Show answer](#)

B. Headache + vomiting + ataxia + posterior fossa mass = cerebellar tumor; medulloblastoma is the most common pediatric CNS tumor and classically presents this way.

Q23. Select all that apply. Which features are common in childhood cancers?

- A. Frequent origin from embryonal tissue (-blastomas)
- B. Often associated with germline predisposition
- C. Tend to be more chemo-responsive than adult cancers
- D. Driven primarily by long-term carcinogen exposure (smoking, alcohol)
- E. Hematopoietic and CNS malignancies are particularly common

[Show answer](#)

A, B, C, E. D is incorrect — childhood cancers are usually NOT driven by accumulated environmental exposures, which is one reason they're biologically different from adult cancers.

Q24. Order. Place these events of neuroblastoma development in correct order:

- Mass effect on adjacent structures and metastasis
- Mutations accumulate in neural crest progenitor cells
- Abnormal proliferation of immature neuroblasts in the adrenal medulla or sympathetic chain
- Tumor enlarges with calcifications and produces excess catecholamines

[Show answer](#)

Correct order: Mutations accumulate in neural crest progenitors → abnormal proliferation in adrenal/sympathetic chain → tumor enlarges with calcifications and catecholamine excess → mass effect and metastasis.

Unit 6 • Adolescent Cancers LO3

6.1 Primary Bone Tumors

Tumor	Age	Location	Imaging clue
Osteosarcoma	10–20 years (peak)	Metaphysis of long bones — distal femur, proximal tibia, proximal humerus	"Sunburst" pattern, Codman triangle (periosteal reaction); destructive lesion
Ewing sarcoma	5–25 years	Diaphysis of long bones; pelvis	"Onion-skin" periosteal reaction; t(11;22) translocation
Osteoid osteoma	10–20 years	Long bones (often femur/tibia)	Small lucent nidus; pain relieved by NSAIDs (benign)

PATHOLOGY Classic osteosarcoma vignette: "Adolescent with persistent knee or upper-arm pain and swelling; imaging reveals destructive lesion of distal femur or proximal

humerus with periosteal reaction." Survivors of hereditary retinoblastoma are at much higher risk of osteosarcoma later — both involve RB1.

6.2 Lymphomas

	Hodgkin lymphoma	Non-Hodgkin lymphoma
Age peak	Bimodal: 15–35 and >55	All ages, increases with age
Spread	Contiguous, predictable from one node group to next	Non-contiguous, often extranodal
Histology	Reed-Sternberg cells (bilobed "owl-eye" cells)	Various — diffuse large B-cell, follicular, Burkitt, etc.
Symptoms	Painless cervical lymphadenopathy, fever, night sweats, weight loss ("B symptoms")	Painless lymphadenopathy; B symptoms in aggressive types
Prognosis	Very good with treatment	Highly variable; depends on subtype

★ **EXAM TIP** "Adolescent with painless neck swelling, fever, and night sweats" = think Hodgkin lymphoma. Confirm with lymph node biopsy looking for Reed-Sternberg cells. Treatment is highly effective.

ACRONYM / MNEMONIC B symptoms: drenching night sweats, fevers, unexplained weight loss > 10% in 6 months. Their presence stages the disease as "B" (vs "A" without them) and signals more aggressive disease.

Soft tissue sarcomas in young patients

- **Rhabdomyosarcoma** — malignant tumor of skeletal muscle origin; most common soft tissue sarcoma in children. Head/neck, GU tract, extremities.
- **Liposarcoma, leiomyosarcoma, fibrosarcoma** — older patients more typically.
- Sarcomas tend to spread **hematogenously** (often to lung) rather than lymphatically.

Practice Questions — Adolescent Cancers

Q25. A 14-year-old has persistent bone pain and swelling in the upper arm. Imaging shows a destructive lesion of the humerus with periosteal reaction. Most likely diagnosis?

- A. Ewing sarcoma
- B. Osteosarcoma
- C. Osteoid osteoma
- D. Multiple myeloma

Show answer

B. Adolescent + metaphysis of long bone (proximal humerus) + destructive lesion with periosteal reaction = osteosarcoma. Ewing typically arises from the diaphysis and shows onion-skin periosteum.

Q26. A 17-year-old has painless neck swelling, fever, night sweats. Most likely diagnosis?

- A. Acute myeloid leukemia
- B. Hodgkin lymphoma
- C. Wilms tumor
- D. Medulloblastoma

[Show answer](#)

B. Painless lymphadenopathy + B symptoms (fever, night sweats, weight loss) in an adolescent = Hodgkin lymphoma. Biopsy shows Reed-Sternberg cells.

Q27. Select all that apply. Which cancers develop from muscle or other soft tissues?

- A. Rhabdomyosarcoma
- B. Leiomyosarcoma
- C. Liposarcoma
- D. Hodgkin lymphoma
- E. Adenocarcinoma

[Show answer](#)

A, B, C. Rhabdomyosarcoma (skeletal muscle), leiomyosarcoma (smooth muscle), liposarcoma (adipose). D is hematopoietic; E is epithelial.

Q28. Select all that apply. Which cancers are blood or immune system malignancies seen in adolescents?

- A. Acute lymphoblastic leukemia
- B. Acute myeloid leukemia
- C. Hodgkin lymphoma
- D. Non-Hodgkin lymphoma
- E. Osteosarcoma

[Show answer](#)

A, B, C, D. E (osteosarcoma) is a bone tumor, not a blood/immune system malignancy.

Appendix • Top Missteps & Master Mnemonics

Top 15 Week 3 Missteps

#	The trap	What's actually true
1	"Oncogene activation needs both alleles."	Only ONE allele needs to be activated (gain of function = dominant). Tumor suppressors need both lost.
2	"All -omas are benign."	Lymphoma, melanoma, hepatoma, seminoma are malignant despite the "-oma" suffix.
3	"Dysplasia is cancer."	Pre-cancerous, often reversible. Carcinoma in situ is the immediate precursor; invasive crosses the basement membrane.
4	"Stage and grade are the same."	Grade = differentiation (microscope). Stage = spread (TNM).
5	"Cachexia is just not eating enough."	

		Cytokine-driven hypermetabolism — proteolysis, lipolysis, anorexia, insulin resistance. Eating more does not fix it.
6	"Wilms and neuroblastoma are the same."	Wilms = kidney, doesn't cross midline. Neuroblastoma = adrenal/sympathetic, often crosses midline, calcifications, ↑ HVA/VMA.
7	"Leukocoria is harmless."	Leukocoria = retinoblastoma until proven otherwise. Urgent referral.
8	"Childhood cancers are caused by smoking."	Children's cancers are largely not driven by environmental carcinogens; they often arise from embryonal tissue with germline factors.
9	"Hodgkin and non-Hodgkin lymphoma are clinically identical."	Hodgkin spreads contiguously, has Reed-Sternberg cells, and has bimodal age distribution.
10	"Osteosarcoma occurs in the diaphysis."	Osteosarcoma = metaphysis (distal femur, proximal tibia, proximal humerus). Ewing = diaphysis.
11	"VEGF blockers cure cancer."	Anti-VEGF therapy slows angiogenesis but is one tool among many; resistance develops.
12	"BRCA1/2 mutations only matter for breast cancer."	Also raise risk for ovarian, prostate, pancreatic, and others.
13	"Acetaldehyde is a harmless byproduct."	It's the carcinogenic intermediate of alcohol metabolism that damages DNA.
14	"DES affects the mother only."	Daughters of DES-exposed mothers have ↑ risk of clear cell vaginal/cervical cancer.
15	"Obesity is a cardiovascular issue only."	Obesity drives endometrial, postmenopausal breast, colorectal, kidney, esophageal, pancreatic, and liver cancers.

Master Mnemonic Sheet

Mnemonic	Meaning
"Gas vs Brake"	Oncogenes are stuck-on gas pedals; tumor suppressors are broken brakes
Knudson 2-hit	Tumor suppressors require both alleles inactivated (recessive at the cell level)
"Stage = Spread, Grade = Grossness"	Stage = TNM (extent). Grade = differentiation (microscopic appearance).
"Big 4 mets"	Most common hematogenous metastasis sites: liver, lung, bone, brain
"-carcinoma vs -sarcoma vs leukemia/lymphoma vs -blastoma"	Epithelial vs connective vs hematopoietic vs embryonal malignancies
Malignant -omas	Lymphoma, melanoma, hepatoma, seminoma — exceptions to "-oma = benign" rule
"Wilms = kidney, doesn't cross midline"	

	vs neuroblastoma which crosses midline and has calcifications
Leukocoria = retinoblastoma	White pupil reflex in young child
"Sunburst metaphysis"	Osteosarcoma imaging
"Onion-skin diaphysis"	Ewing sarcoma imaging
Reed-Sternberg = Hodgkin	Bilobed "owl-eye" cells on biopsy
B symptoms	Drenching night sweats, fever, weight loss > 10% in 6 months
Acetaldehyde = alcohol's bad metabolite	Driver of alcohol-related carcinogenesis
HPV E6 = p53, E7 = Rb	How HPV inactivates two tumor suppressors

Pediatric Cancer Quick Reference

Pattern	Cancer
4yo with fatigue, fever, petechiae, blasts on CBC	ALL
3yo with painless unilateral abdominal mass, hematuria	Wilms tumor
2yo with adrenal mass + calcifications, ↑ HVA/VMA	Neuroblastoma
Toddler with white pupil reflex	Retinoblastoma
Child with morning headache, vomiting, ataxia	Cerebellar tumor (medulloblastoma)
Adolescent with knee pain, distal femur lesion	Osteosarcoma
Teen with diaphyseal bone lesion, onion-skin periosteum	Ewing sarcoma
Teen with painless cervical lymphadenopathy + B symptoms	Hodgkin lymphoma

Week 3 Study Strategy

- **Drill the gas-vs-brake gene framework first.** Half the LO1 questions decompose to oncogene vs tumor suppressor logic.
- **Memorize the development cascade.** Hyperplasia → dysplasia → CIS → invasive → metastatic. Basement membrane crossing is the milestone.
- **Understand cachexia is cytokine-driven, not anorexia.** SATA traps will include "the patient is just not eating." That's wrong.
- **Pair every risk factor with its mechanism.** Smoking → carcinogens + metaplasia. Alcohol → acetaldehyde. Obesity → inflammation + insulin + estrogen.
- **Match each pediatric tumor to its single best diagnostic clue.** Crossing the midline (neuroblastoma), confined to kidney (Wilms), white pupil (retinoblastoma), posterior fossa ataxia (medulloblastoma).
- **Know which "-omas" are malignant.** Lymphoma, melanoma, hepatoma, seminoma. Easy points if you have it.
- **For adolescent cancers, lock down osteosarcoma vs Ewing and Hodgkin lymphoma's classic vignette.**

SCI225 Week 3 Study Guide — built from the Week 3 learning objectives covering cancer biology, tumor behavior, naming/grading/staging, cancer epidemiology, and pediatric and adolescent cancers. Made by Dr. Vidya Aggrawal to teach the concepts tested, not the exam questions themselves. Same color-coded format as the Week 1, 2, 4, and 5 guides. Print for paper, or use in-browser with collapsible practice answers.